SMART INDIA HACKATHON 2026

SMART INDIA HACKATHON 2026

TITLE PAGE

VYUHA

Adaptive counter-drone decision training
Train. Explain. Adapt. Repeat.

Problem Statement ID – SIH26247

Problem Statement Title –

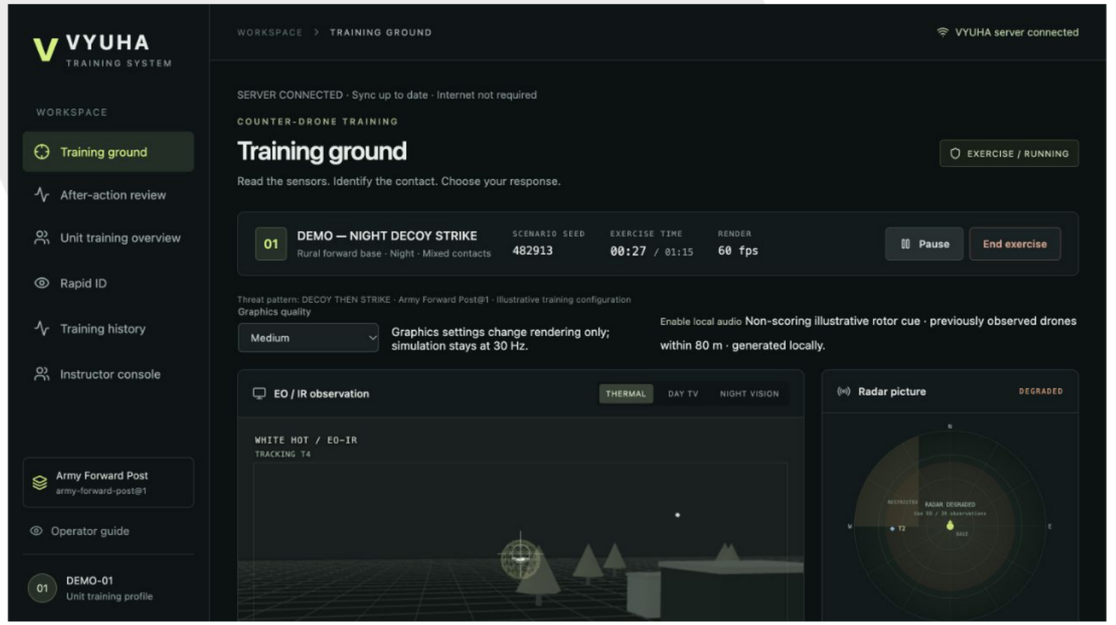
AI-Enabled Drone & Counter-Drone Threat Simulation Trainer

Theme – Robotics and Drones

PS Category – Software

Team ID – 185350

Team Name (Registered on portal) – WhiteList&0



WhiteList
&0

IDEA TITLE

SMART INDIA HACKATHON 2026

Proposed Solution (Describe your Idea/Solution/Prototype)

Detailed explanation of the proposed solution

A local laptop serves repeatable mixed-contact drills; verified decisions feed explainable review, persistent skill profiles and the next adaptive exercise.

How it addresses the problem

PS requirement	Working VYUHA capability
Recognise / classify / respond	Tactical drills + separate Rapid ID
Scoring / after-action review	Server reconstruction + Ghost + Time Machine
Adjust difficulty	Four-skill Beta Director; heuristic difficulty
Conditions / terrain	Day/night, sensor degradation, urban/rural
Single / swarm / unit training	Five Red Force patterns + local LAN Unit Mode

Innovation and uniqueness of the solution

One traceable loop: issued scenario → decisions → verified AAR → profile → next drill.

WhiteList &0

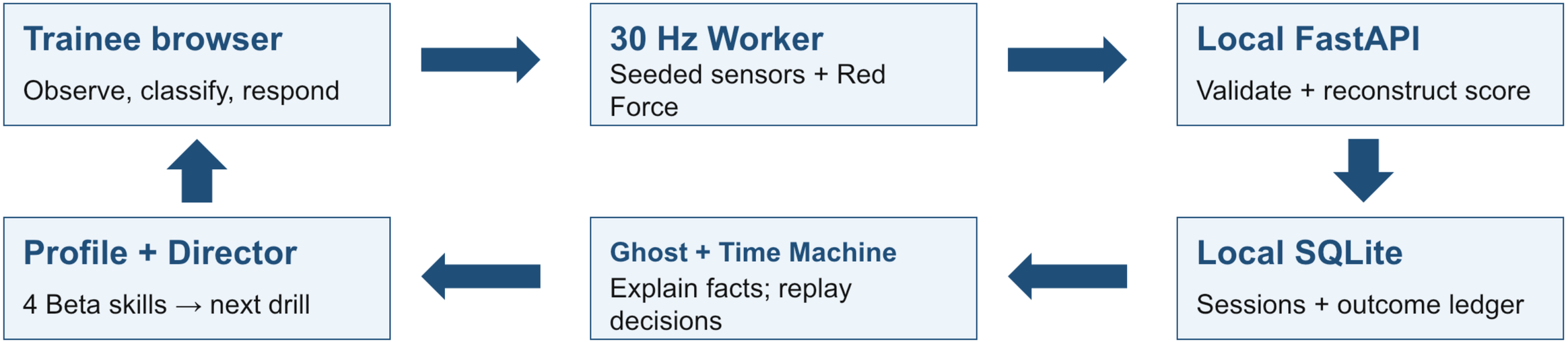
TECHNICAL APPROACH

SMART INDIA HACKATHON 2026

Technologies to be used (e.g. programming languages, frameworks, hardware)

React + TypeScript + Three.js · Web Worker · Python/FastAPI · SQLite + IndexedDB

Methodology and process for implementation (Flow Charts/Images/ working prototype)



Detect	Classify	Prioritise	Engage
25 points	25 points	15 points	35 points

WhiteList &0

FEASIBILITY AND VIABILITY

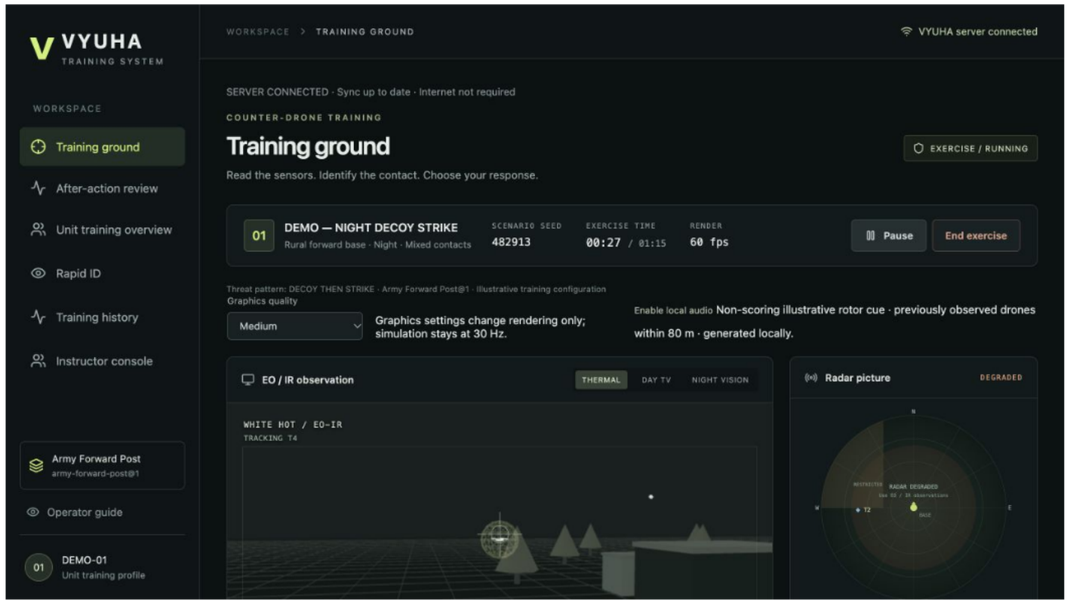
SMART INDIA HACKATHON 2026

Analysis of the feasibility of the idea

134 backend · 44 frontend · 312 parity checkpoints · 22 local assets

Physical multi-device LAN training was validated. Runtime assets and services are hosted locally; dedicated WAN-disconnected physical validation remains pending.

M1 Air / 8 GB: median 60 FPS at 8–200 contacts, fixed Medium, 20-second runs. Six simulated clients / 18 completions over loopback; physical Mac + Android LAN.



Potential challenges and risks

Trusted LAN, no production authentication. Illustrative laws; no human efficacy study.

Strategies for overcoming these challenges

Local assets, queued recovery, backup/restore; Tactical 2D and recorded-demo fallbacks.

Physical WAN-off/LAN-on: NOT PERFORMED BEFORE SUBMISSION
Reason: Required physical/network setup unavailable during final validation window.

WhiteList &0

IMPACT AND BENEFITS

SMART INDIA HACKATHON 2026

Potential impact on the target audience

Repeat mixed-contact decisions safely.
Review the evidence behind each mistake.
Select the next drill from saved outcomes.

Benefits of the solution (social, economic, environmental, etc.)

Intended: accessible repetition on unit laptops and traceable instructor review. Cost savings and field-performance transfer are unmeasured.

SIMULATED ONLY · not human efficacy

Metric	Director	Random
Reached specified threshold	8 / 50	5 / 50
Median sessions, reached only	21.5	28
Mean final hidden probability	73.43%	73.15%

50 paired profiles per policy × 30 sessions.
Abstract Bernoulli learning law, uncalibrated; no human learners or tactical physics.

Human efficacy has not been measured.

Next: subject-matter calibration and a consented pre/post pilot. Evaluation tooling is experimental.

WhiteList &0

RESEARCH AND REFERENCES

SMART INDIA HACKATHON 2026

Details / Links of the reference and research work

Official problem statement and 2026 submission template

sih.gov.in/sih2026PS · sih.gov.in/letters/2026/SIH2026-IDEA-Presentation-Format.pptx

Russo et al. — A Tutorial on Thompson Sampling (2018)

arxiv.org/abs/1707.02038 · Method reference; does not validate VYUHA training efficacy.

Implementation references

developer.mozilla.org/en-US/docs/Web/API/Web_Workers_API/Using_web_workers
fastapi.tiangolo.com/tutorial/static-files/

Reproducible product evidence

FINAL-VALIDATION.md · release-evidence.json · feature-manifest.json
Private repository: github.com/vishruthvayu/vyuha-sih26247 · source package supplied separately.

DEFERRED: calibrated signatures · human/field evaluation · production identity · live feeds · crew · VR · rain